Ma 3 PS1

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§1 Two chess players, A and B, are going to play 7 games. Each game has three possible outcomes: a win for A (which is a loss for B), a draw (tie), and a loss for A (which is a win for B). A win is worth 1 point, a draw is worth 0.5 points, and a loss is worth 0 points.

§1.1 How many possible outcomes for the individual games are there, such that overall player A ends up with 3 wins, 2 draws, and 2 losses?

We can represent the match as a string of length 7 over the alphabet $\{W, D, L\}$. We want strings with exactly 3 W's, 2 D's, and 2 L's. The number of such strings is the number of permutations of a multiset with multiplicities $(3, 2, 2)$:

$$\frac{7!}{3!\,2!\,2!}.$$

$$\boxed{\frac{7!}{3!\,2!\,2!} = 210}$$

§1.2 How many possible outcomes for the individual games are there, such that A ends up with 4 points and B ends up with 3 points?

Given the described record and points system,

$$w + d + \ell = 7, \qquad w + \frac{1}{2}d = 4.$$

Multiplying the points equation by 2 gives $2w + d = 8$, so $d = 8 - 2w$. Substituting into $w + d + \ell = 7$ gives

$$\ell = 7 - w - d = 7 - w - (8 - 2w) = w - 1.$$

Now $w \in \{1, 2, 3, 4\}$ gives the possibke triples

$$(1, 6, 0),\ (2, 4, 1),\ (3, 2, 2),\ (4, 0, 3).$$

For each triple (w, d, ℓ), we do the same as in 1.1:

$$\frac{7!}{1!\,6!\,0!} + \frac{7!}{2!\,4!\,1!} + \frac{7!}{3!\,2!\,2!} + \frac{7!}{4!\,0!\,3!} = 7 + 105 + 210 + 35 = 357.$$

$$\boxed{357}$$

§1.3 Now assume that they are playing a best-of-7 match, where the match will end when either player has 4 points or when 7 games have been played, whichever is first. For example, if after 6 games the score is 4 to 2 in favor of A, then A wins the match and they don't play a 7th game. How many possible outcomes for the individual games are there, such that the match lasts for 7 games and A wins by a score of 4 to 3?

For the match to last all 7 games, neither player can have reached 4 points after 6 games. The last game cannot be a loss for A (that would decrease A's final score), so game 7 is either a win or a draw.

Case 1: Game 7 is a win for A. Then after 6 games the score must be 3–3. Let (w, d, ℓ) be counts in the first 6 games; then

$$w + d + \ell = 6, \qquad w + \frac{1}{2}d = 3 \iff 2w + d = 6.$$

The solutions are $(0, 6, 0), (1, 4, 1), (2, 2, 2), (3, 0, 3)$. Number of cases are

$$\frac{6!}{0!\,6!\,0!} + \frac{6!}{1!\,4!\,1!} + \frac{6!}{2!\,2!\,2!} + \frac{6!}{3!\,0!\,3!} = 1 + 30 + 90 + 20 = 141.$$

Case 2: Game 7 is a draw. Then after 6 games the score must be 3.5–2.5. For the first 6 games,

$$w + d + \ell = 6, \qquad w + \frac{1}{2}d = 3.5 \iff 2w + d = 7.$$

The solutions are $(1, 5, 0), (2, 3, 1), (3, 1, 2)$, contributing

$$\frac{6!}{1!\,5!\,0!} + \frac{6!}{2!\,3!\,1!} + \frac{6!}{3!\,1!\,2!} = 6 + 60 + 60 = 126.$$

Adding the two cases gives

$$141 + 126 = \boxed{267}$$

§2 Four players named A, B, C, and D, are playing a card game. A standard well-shuffled deck of cards is dealt to the players (so each player receives a 13-card hand).

§2.1 (a) How many possibilities are there for the hand that player A will get?

A's hand is a 13 card subset of the 52 card deck so the number of possible hands is

$$\boxed{\binom{52}{13}.}$$

§2.2 (b) How many possibilities are there overall for what hands everyone will get, assuming that it matters which player gets which hand, but not the order of cards within a hand?

First choose A's 13 cards: $\binom{52}{13}$. Then choose B's 13 from the remaining 39: $\binom{39}{13}$. Then choose C's 13 from the remaining 26: $\binom{26}{13}$. The remaining 13 cards automatically go to D: $\binom{13}{13} = 1$.

So the total number of possible (labeled) deals is

$$\boxed{\binom{52}{13}\binom{39}{13}\binom{26}{13}\binom{13}{13}}$$

§2.3 (c) Explain intuitively why the answer to Part (b) is not the fourth power of the answer to Part (a).

If we took $\left(\binom{52}{13}\right)^4$, we would be choosing a 13 card hand for A, then independently choosing a 13 card hand for B from the full deck again, and same for C and D. This allows the same card to appear in multiple players hands.

$$\boxed{\text{Because the four hands must be disjoint and exhaust the deck, the choices are not independent.}}$$

§3 Give a story proof that $\sum_{k=2}^{n} \binom{k}{2}\binom{n-k+2}{2} = \binom{n+3}{5}$ for all positive integers n. Hint: Consider the middle number in a subset of $1, 2, \ldots, n+3$ of size 5.

We can count the number of 5 element subsets of $\{1, 2, \ldots, n+3\}$ in two ways.

Right side: Choose 5 elements directly from $\{1, 2, \ldots, n+3\}$. There are $\binom{n+3}{5}$ ways.

Left side: Organize by the middle element. Let the 5 chosen elements in increasing order be $a < b < c < d < e$. The middle element is c.

Since there are exactly 2 elements below c and 2 above c, we have $c \in \{3, 4, \ldots, n+1\}$.

Let $k = c - 1$, so $k \in \{2, 3, \ldots, n\}$ and $c = k + 1$. For each value of k:

- Choose 2 elements from $\{1, 2, \ldots, k\}$: $\binom{k}{2}$ ways
- Choose 2 elements from $\{k+2, k+3, \ldots, n+3\}$: $\binom{n-k+2}{2}$ ways

By the multiplication principle:

$$\sum_{k=2}^{n} \binom{k}{2}\binom{n-k+2}{2}$$

Both methods count the same 5-element subsets, so:

$$\boxed{\sum_{k=2}^{n} \binom{k}{2}\binom{n-k+2}{2} = \binom{n+3}{5}} \qquad \square$$

§4 A deck of cards is shuffled well. The cards are dealt one by one, until the first time an ace appears.

§4.1 (a) Find the probability that no kings, queens or jacks appear before the first ace.

The event no kings, queens, or jacks appear before the first ace means that among the 16 total relevant cards, the first one is an ace. By symmetry, each of the 16 distinguished cards is equally likely to be the earliest, and 4 of them are aces. So

$$\boxed{\mathbb{P}(\text{no K/Q/J before first ace}) = \frac{4}{16} = \frac{1}{4}.}$$

§4.2 (b) Find the probability that exactly one king, exactly one queen and exactly one jack appear (in any order) before the first ace.

We can sumulate the process where one king, queen, and jack appear (in any order) before the first ace in the same 16 card scenario as above.

- For the first car selection, we can pick a king, queen, or jack, or which the probability is $\frac{12}{16}$.
- For the second card selection, we can pick any card of the two remaining face cards of which the probability is $\frac{8}{15}$ as there are 4 of each face cards in the deck and 15 cards remaining.
- For the third card selection, we can pick any card of the one remaining face cards of which the probability is $\frac{4}{14}$ as there are 4x of our final face card in the deck and 14 cards remaining.
- Finally, for the selection of the Ace, we have 4 possible cards and 13 cards remaining, yielding a probability of $\frac{4}{13}$.

Using multiplication rule:

$$\mathbb{P}(\text{exactly one K, one Q, one J before first ace}) = \frac{12}{16} \cdot \frac{8}{15} \cdot \frac{4}{14} \cdot \frac{4}{13} = \boxed{\frac{16}{455}}$$

§5 Let A and B be events. The symmetric difference $A \triangle B$ is defined to be the set of all elements that are in A or B but not both (XOR). Show that $\mathbb{P}(A \triangle B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B)$.

By definition,

$$A \triangle B = (A \setminus B) \cup (B \setminus A),$$

and the two sets $(A \setminus B)$ and $(B \setminus A)$ are disjoint. This means

$$\mathbb{P}(A \triangle B) = \mathbb{P}(A \setminus B) + \mathbb{P}(B \setminus A).$$

Also,

$$A = (A \setminus B) \cup (A \cap B) \quad \text{(disjoint union)},$$

so $\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$. Similarly, $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Substituting,

$$\mathbb{P}(A \triangle B) = \big(\mathbb{P}(A) - \mathbb{P}(A \cap B)\big) + \big(\mathbb{P}(B) - \mathbb{P}(A \cap B)\big) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).$$

$$\boxed{\mathbb{P}(A \triangle B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).}$$

§6 A club consists of 15 seniors, 10 juniors, and 12 sophomores. An organizing committee of size 5 is chosen randomly (with all subsets of size 5 equally likely).

There are $15 + 10 + 12 = 37$ students total, so the sample space has size $\binom{37}{5}$.

§6.1 Find the probability that there are exactly 3 seniors in the committee.

Choose 3 of the 15 seniors and the remaining 2 members from the 22 non-seniors (juniors+sophomores):

$$\mathbb{P}(\text{exactly 3 seniors}) = \frac{\binom{15}{3}\binom{22}{2}}{\binom{37}{5}}.$$

$$\boxed{\frac{\binom{15}{3}\binom{22}{2}}{\binom{37}{5}}.}$$

§6.2 Find the probability that the committee has at least one representative from each of the senior, junior, and sophomore classes.

Let E be the event that the committee has at least one senior, junior, and sophomore. Use the complement:

$$\mathbb{P}(E) = 1 - \mathbb{P}(E^c),$$

where E^c is the event that at least one class is missing.

Let

$$A = \{\text{no seniors}\}, \quad B = \{\text{no juniors}\}, \quad C = \{\text{no sophomores}\}.$$

Then $E^c = A \cup B \cup C$, so by inclusion–exclusion,

$$\mathbb{P}(E^c) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C),$$

since $A \cap B \cap C$ is impossible for a size-5 committee.

Count each term by choosing all 5 committee members from the allowed pool:

$$\mathbb{P}(A) = \frac{\binom{22}{5}}{\binom{37}{5}}, \quad \mathbb{P}(B) = \frac{\binom{27}{5}}{\binom{37}{5}}, \quad \mathbb{P}(C) = \frac{\binom{25}{5}}{\binom{37}{5}},$$

$$\mathbb{P}(A \cap B) = \frac{\binom{12}{5}}{\binom{37}{5}}, \quad \mathbb{P}(A \cap C) = \frac{\binom{10}{5}}{\binom{37}{5}}, \quad \mathbb{P}(B \cap C) = \frac{\binom{15}{5}}{\binom{37}{5}}.$$

Therefore,

$$\mathbb{P}(E) = 1 - \frac{\binom{22}{5} + \binom{27}{5} + \binom{25}{5} - \binom{12}{5} - \binom{10}{5} - \binom{15}{5}}{\binom{37}{5}}.$$

$$\boxed{\mathbb{P}(E) = \frac{93250}{145299} \approx 0.64178}$$

§7 Suppose A and B are independent events. Show that A^c and B^c are also independent.

Since A and B are independent,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

We also know

$$A^c \cap B^c = (A \cup B)^c \quad \Rightarrow \quad \mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B).$$

Also,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B).$$

Therefore,

$$\begin{aligned}\mathbb{P}(A^c \cap B^c) &= 1 - \big(\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B)\big) \\ &= \big(1 - \mathbb{P}(A)\big)\big(1 - \mathbb{P}(B)\big) \\ &= \mathbb{P}(A^c)\mathbb{P}(B^c).\end{aligned}$$

Thus means A^c and B^c are independent.

$$\boxed{\mathbb{P}(A^c \cap B^c) = \mathbb{P}(A^c)\mathbb{P}(B^c)}$$

§8 Two cards are distributed to each of three players from a standard deck of 52 shuffled cards. What is the probability that at least two of the three players have an ace with either a jack or a king?

Let E_i be the event that player i's 2 card hand satisfied the problem's conditions. We can now divide the problem into subparts.

¶ Total outcomes. Dealing 2 cards to each labeled player with order within each hand ignored yields

$$N = \binom{52}{2}\binom{50}{2}\binom{48}{2}.$$

¶ Inclusion-exclusion for at least two. "At least two of E_1, E_2, E_3" means $(E_1 \cap E_2) \cup (E_1 \cap E_3) \cup (E_2 \cap E_3)$. Applying inclusion-exclusion to this union with $A = E_1 \cap E_2$, $B = E_1 \cap E_3$, $C = E_2 \cap E_3$:

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ + \mathbb{P}(A \cap B \cap C)\end{aligned}$$

Substituting back:

$$\begin{aligned}&= \mathbb{P}(E_1 \cap E_2) + \mathbb{P}(E_1 \cap E_3) + \mathbb{P}(E_2 \cap E_3) \\ &\quad - \mathbb{P}(E_1 \cap E_2 \cap E_3) - \mathbb{P}(E_1 \cap E_2 \cap E_3) - \mathbb{P}(E_1 \cap E_2 \cap E_3) \\ &\quad + \mathbb{P}(E_1 \cap E_2 \cap E_3) \\ &= \sum_{i<j} \mathbb{P}(E_i \cap E_j) - 2\mathbb{P}(E_1 \cap E_2 \cap E_3)\end{aligned}$$

By symmetry, all pairwise probabilities are equal:

$$\mathbb{P}(\text{at least two of } E_1, E_2, E_3) = 3\mathbb{P}(E_1 \cap E_2) - 2\mathbb{P}(E_1 \cap E_2 \cap E_3).$$

¶ Count $E_1 \cap E_2$. There are 4 aces and 8 relevant face cards (4 jacks + 4 kings).

- Player 1: choose 1 ace and 1 face card: $4 \cdot 8$ ways.
- Player 2: choose from the remaining cards: $3 \cdot 7$ ways.
- Player 3: any 2 of the remaining 48 cards: $\binom{48}{2}$ ways.

So

$$|E_1 \cap E_2| = (4 \cdot 8)(3 \cdot 7)\binom{48}{2}.$$

¶ Count $E_1 \cap E_2 \cap E_3$. Similarly,

$$|E_1 \cap E_2 \cap E_3| = (4 \cdot 8)(3 \cdot 7)(2 \cdot 6).$$

¶ Anwser

$$\boxed{\mathbb{P}(\text{at least two}) = \frac{3(4 \cdot 8)(3 \cdot 7)\binom{48}{2} - 2(4 \cdot 8)(3 \cdot 7)(2 \cdot 6)}{\binom{52}{2}\binom{50}{2}\binom{48}{2}} = \frac{64}{51935} \approx 0.001232.}$$